Zexin Jian

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My research interest includes:

- Managed Runtime System
- Machine Learning System
- Large Language Models (LLM)

I will continue to search, even if the endless stars make my search hopeless, even if I am alone.

Education Background

Sep.2021 ~ present
Changsha, China
National University of Defense Technology (NUDT)
B.E. Student, School of Computer Science
Advised by Dr. Zengkun

Sep.2025 ~ present
Beijing, China
Chinese Academy of Sciences (CAS)
ph.D. Student, Institute of Computing Technology (ICT)
co-Advised by Dr. Huimin Cui and Dr. Chenxi Wang

Research & Publications

- **[NeurIPS'24 CCF-A]** Zhihua Wen, Zhiliang Tian, **Zexin Jian**, Zhen Huang, Pei Ke, Yifu Gao, Minlie Huang, Dongsheng Li. *Perception of Knowledge Boundaries for LLMs via Semi-open-ended Question Answering*.
- **[ACL'23 CCF-A]** Zhihua Wen, Zhiliang Tian, Zhen Huang, Yuxin Yang, **Zexin Jian**, Changjian Wang, Dongsheng Li. *GRACE: Gradient-guided Controllable Retrieval for Augmenting Attribute-based Text Generation*.

Projects

Sidecar: Megakernel Compiler and Runtime for Ascend NPUs **Co-lead**
Noah's Ark Lab, Huawei | Ascend C/C++, Python, CANN
Co-led development of a persistent megakernel system for LLM decoding, addressing repeated kernel launches, operator-wide synchronization, and scheduling overhead across heterogeneous Cube/Vector units.

- **Task-graph compilation:** The compiler derives tile dependencies from tensor access regions to enable cross-operator pipelining. Transitive-edge pruning and virtual synchronization nodes reduce dependency edges by **86.29%** on Qwen3-30B-A3B (batch size 1), while preserving execution dependencies.
- **Heterogeneous scheduling:** The AICPU control plane combines per-worker HBM task queues with register-based notifications, supporting Cube, Vector, and coordinated Mix tasks. Parallel schedulers, batched dispatch, and runtime MoE expert filtering reduce control overhead and balance ready work across compatible workers.
- **Scheduling-computation overlap:** Early handoff separates scheduling readiness from data readiness, preparing successors during producer execution while worker-side counters protect dependent accesses. This mechanism reduces time per output token (TPOT) by **5.98-11.77%** in GLM-5.2 W4A8 ablations (TP=8).
- **End-to-end results:** Evaluated across Ascend **A2/A3/A5** on dense and MoE models, the system achieves lower TPOT in **all 42 configurations** than both optimized vLLM-Ascend and SGLang-Ascend. Peak TPOT reduction is **40.72%** versus the faster baseline (**34.23 ms to 20.29 ms**, GLM-5.2 W4A8, A5, TP=8, batch size 1).

ServerlessAI **Noah's Ark Lab, Huawei** *2025-12~2026-12*
CloudMatrix384

AI Compiler for RISC-V CIM **AIIT, PKU** *2025-05~2028-01*
C/C++, MLIR, RISCV, Compiler Design

- Developed using C/C++, this compiler system, built on MLIR, enables automatic back-end optimization for various RISC-V SRAM computing-in-memory chips
- The project's starting point is to translate applications to this unified intermediate representation, perform application feature analysis on this representation, identify computationally accelerable parts, and translate them into specific RISC-V acceleration instructions, fully leveraging the high-performance features of the RISC-V general-purpose core and NPU acceleration core

LLM as Compiler **ICT, CAS** *2024-09~2025-04*

LLM, Compiler Design, Machine Learning System

- Responsible for designing and optimizing the compiler module within the Large Processor Chip Model (LPCM) framework, enabling automatic conversion from source code to hardware-adaptive machine code
- Successfully automated the compilation process by introducing a Large Language Model (LLM), supporting hardware-software co-design. This has driven multi-level optimization of compilers from human-machine collaboration to automation

mini RustOS **NUDT** *2024-09~2024-12*

Rust, X86, OS Design, Hybrid Kernel

- Developed an experimental operating system kernel for the X86 architecture using Rust as the primary programming language.
- Starting from the macro kernel, absorbing and learning from the design patterns of the micro kernel to improve security while ensuring certain performance.

SysY Compiler **NUDT** *2024-02~2024-08*

C/C++, LLVM, RISCV, Compiler Design, Multi-Threads Parallel

- Developed using C/C++, implementing a fully functional compiler system for SysY2022 (a subset of C)
- Key tasks included front-end lexical and semantic analysis, as well as low-level code generation and optimization for the RISC-V architecture (including code generation and register allocation)

GROVE **NUDT** *2023-05~2023-10*

Python, LLM, Controllable Text Generation

- Led the development of the **GROVE retrieval-augmented story generation framework**, integrating large language models, conditional retrieval, and iterative reasoning to generate narratives with complex plot structures
- Designed the novel **Asking-Why prompting mechanism** to construct evidence chains for completing background reasoning and improving logical consistency
- Achieved state-of-the-art results across multiple evaluation metrics, significantly enhancing model **complexity, creativity, and controllability** in story generation tasks

Expertise

Programming	C/C++, Python, Rust
Skills	Linux, PyTorch, CMake, CUDA